A Differential Equation Model for Apple Orchard Yield

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Abstract

This paper develops a mathematical model for fruit yield at a hypothetical apple orchard in the Pacific Northwest, where the majority of for-market apples in the United States are grown. The model considers the effects of soil moisture, temperature, labor availability, and pest pressure on apple production. A system of nonlinear ordinary differential equations is used to describe the intermingled dynamics of fruit load and soil moisture, incorporating external forcing via environmental and operational (i.e. labor) factors.

Analysis of the model finds the existence of two equilibrium points, one corresponding to a zero-fruit-yield equilibrium and subsequent orchard collapse, and the second of which is a positive equilibrium correlated with sustained fruit production. Stability analysis shows that the zero-yield equilibrium may be stable or unstable depending on parameter values, while the positive equilibrium, when it exists, is locally asymptotically stable. Additionally, the model predicts a threshold condition in which orchard viability relies on whether effective growth exceeds total loss pressure.

Numerical simulations of the model demonstrate the impact of both environmental and operational stressors, such as freeze events and labor shortages, on both soil moisture and fruit production at the imagined Washington-state orchard. These results provide insight into the conditions ideal for sustainable orchard yield as well as shed light on the importance of external stress factors in fruit-bearing agricultural systems.

1 Introduction

The American Pacific Northwest, and more specifically Washington state, accounts for over 78 percent of apple production in the United States, with orchards spanning over 178,000 acres [1]. This has significant implications for the apple market, both for farmers and for entities such as impact strategy investment firms that are making a shift to investing in sustainable agriculture as means for investor revenue [2]. Imperative to the success of these orchards and their viability are environmental conditions out of control to those operating such agricultural endeavors, such as weather and deep freezes in the spring, pest damage to trees and fruit, soil conditions, available seasonal labor, and more. Being attuned to the impact of such factors can help farmers and investment firms alike be more aware of what fruit yield outcome to expect and to therefore adjust their financial expectations accordingly. Creating a system for such awareness is what this model aims to do.

2 Model Formulation

The differential equation model of apple orchard yield considers a host of factors that impact fruit production over time and wraps them into a robust model that can predict orchard viability based on these factors. With the model comes a set of key assumptions. Such assumptions are as follows:

- The orchard is treated as spatially uniform.
- Weather inputs affect the orchard at a total-area scale.
- Soil moisture can be represented by a single average quantity.
- Pest pressure may be aggregated into one variable.
- Labor availability affects orchard management effectiveness and harvest efficiency, and therefore apple yield.
- Fruit growth depends on available water and is reduced by stress factors.
- All parameters are constant over the modeled season unless deliberately varied in simulation.
- The model describes one growing system.
- Fruit growth is a continuous process.

The model arrives at a system of ordinary differential equations that refer to rate of change of fruit yield and rate of change of soil moisture. The equations are as follows:

$$\frac{dF}{dt} = r\left(\frac{M}{K_M + M}\right)\phi(T)F - \alpha P(t)F - \beta\ell(L(t))F - \gamma S(T)F, \tag{1}$$

$$\frac{dM}{dt} = R(t) + I(t) - aFM - bTM - cM. \tag{2}$$

The tables below specify the definition of each variable and/or constant.

Fruit Equation

Term	Meaning	Interpretation
$\frac{dF}{dt}$	Rate of change of fruit yield	How fruit production evolves over time
r	Growth rate constant	Maximum potential growth rate of fruit
$\frac{M}{K_M+M}$	Moisture saturation function	Fruit increases depending on moisture availability but levels off, i.e. diminishing returns
$r\left(\frac{M}{K_M+M}\right)\phi(T)F$	Growth term	Fruit increases depending on moisture availability and temperature conditions
$\phi(T)$	Temperature support function	Captures how favorable a temperature is for growth
$-\alpha P(t)F$	Pest damage term	Fruit loss due to pests
α	Pest impact coefficient	Strength of pest-related damage
$-\beta\ell(L(t))F$	Labor shortage term	Larger when labor is low, smaller when labor is high
$\ell(L(t))$	Labor penalty function	Larger when labor is low, smaller when labor is high
β	Labor impact coefficient	Strength of labor-related effects
$-\gamma S(T)F$	Temperature damage term	Fruit loss due to extreme temperatures (e.g. freeze, extreme heat conditions)
$S(T)$	Temperature stress function	Measures impact of temperature on fruit production
γ	Temperature damage coefficient	Strength of temperature-related damage

Moisture Equation

Term	Meaning	Interpretation
$\frac{dM}{dt}$	Rate of change of soil moisture	How soil moisture evolves over time
$R(t)$	Rainfall input	Natural water added to soil via rainfall
$I(t)$	Irrigation input	Water artificially added to soil via irrigation
$R(t)+I(t)$	Total water input	Combined water supply to orchard
$-aFM$	Crop water uptake	Water absorbed by fruit growth
a	Uptake coefficient	Strength of apple tree water consumption
$-bTM$	Temperature-driven water evaporation	Moisture loss increases with temperature
$-cM$	Natural moisture loss	Drainage and baseline water evaporation
c	Baseline loss rate	Constant moisture decay
b	Temperature loss coefficient	Sensitivity of moisture loss to temperature

Each term in the system is representative of a unique physical or environmental factor affecting fruit yield and soil moisture, allowing the model to capture the nuances of both biological growth processes and external stress factors.

Based on the above, model interpretation is as follows. Rate of change of fruit yield is positively influenced by moisture levels in the orchard soil, as well as influenced by a baseline growth rate constant, and is harmed by pest presence, seasonal labor shortages, and extreme temperature events. Rate of change of soil moisture is influenced positively by rainfall and irrigation, and harmed by crop water uptake (assumed to be less than total water input into the system), evaporation of water from the orchard driven by temperature, and a baseline loss rate of moisture from the soil.

3 Equilibrium and Stability Analysis

To solve the system of ordinary differential equations analytically, the model is simplified through freezing external inputs, collapsing fruit-loss modifiers into constants, and reformulating the equations with these new parameters.

The equation for rate of change of fruit yield becomes

$$\frac{dF}{dt} = \left(A\left(\frac{M}{K_M + M}\right) - B\right)F, \tag{3}$$

where A is now equivalent to the growth term in the above table, while B aggregates all loss terms for fruit yield. F remains equivalent to fruit load and M to soil moisture.

The equation for rate of change of soil moisture becomes

$$\frac{dM}{dt} = U - aFM - dM, \tag{4}$$

where U is equivalent to rainfall and irrigation input combined and d aggregates all loss terms for soil moisture. F remains equivalent to fruit load and M to soil moisture.

Through analytical derivation, two equilibrium points are found:

$$E_1 = \left(0, \frac{U}{d}\right), \tag{5}$$

$$E_2 = \left(\frac{U(A-B) - dBK_M}{aBK_M}, \frac{BK_M}{A-B}\right). \tag{6}$$

Investigation of E_1 reveals a fruit-free equilibrium where fruit yield is zero and orchard production is thus unsustainable, and soil moisture settles at a balance between water input versus moisture loss. In essence, while there is still water in the system, it is not sufficient to sustain fruit production against all other harming factors that have been previously introduced.

The stability of E_1 is determined after the derivation of the Jacobian matrix for the simplified system of ordinary differential equations for rate of change of fruit yield and rate of change of soil moisture. The Jacobian matrix is as follows:

$$J(F, M) = \begin{pmatrix} A\dfrac{M}{K_M + M} - B & AF\dfrac{K_M}{(K_M + M)^2} \\ -aM & -aF - d \end{pmatrix}. \tag{7}$$

After substituting F^* and M^* for F and M in the Jacobian matrix, the eigenvalues for E_1 are determined to be

$$\lambda_1 = A\left(\frac{U}{dK_M + U}\right) - B, \qquad \lambda_2 = -d. \tag{8}$$

As $-d < 0$, it becomes essential to understand what happens when $A\left(\frac{U}{dK_M+U}\right) < B$ and $A\left(\frac{U}{dK_M+U}\right) > B$. This introduces a threshold condition, and compares effective fruit growth supported by soil moisture against the impact of combined pressure loss on fruit growth. In the former case, both eigenvalues for E_1 are negative, so E_1 is stable and fruit growth cannot recover from small perturbations, leading to total orchard collapse and an unviable agricultural operation. In the latter case, one eigenvalue is negative ($-d < 0$) and the other is positive, thus indicating that E_1 is a saddle point. This signifies that in some directions on the phase plane, the system moves towards E_1, while in others it moves away. In other words, under certain conditions, the orchard will be able to recover from collapse and sustain fruit production, while under others, it will collapse.

A look at E_2 identifies it as a positive equilibrium, where fruit yield is positive, soil moisture is positive, growth and losses are balanced, and the orchard is producing fruit while soil moisture stabilizes at a level that supports such production. This is the desired state of the model. However, said positive equilibrium only exists under certain conditions; for E_2 to exist in a manner that is biologically sound, the formulas for coordinates F^* and M^* must be positive. For M^*, this implies $BK_M/(A - B)$ must be greater than zero. Since B and K_M are positive, it follows that $(A - B)$ must be positive, meaning that effective growth strength must exceed pressure loss. For F^*, this implies $(U(A - B) - dBK_M)/(aBK_M)$ must be greater than zero. As a, B, and $K_M > 0$, it follows that $U(A - B) - dBK_M$ must be greater than 0. This means that not only must growth outweigh loss (pests, labor shortages, freezes, etc.), but water input must be strong enough relative to soil moisture loss.

A probe into the eigenvalues for E_2 demonstrates that, as the trace of the Jacobian matrix with M and F substituted with the corresponding F^* and M^* of E_2 is less than zero while the determinant is greater than zero, E_2 is locally asymptotically stable. Once the orchard is near its sustainable state, it tends to return to it despite any small disturbances to the system.

Below is a graph of the phase plane of the reduced system (see Figure 1).

At any point on the graph (F, M), the shown arrows indicate the direction the system moves based on dF/dt and dM/dt. Arrows moving to the right represent fruit yield increasing while arrows to the left represent fruit yield decreasing, and arrows pointing up represent soil moisture increasing while arrows pointing down represent soil moisture decreasing. Each arrow essentially serves as an indication of what to expect will unfold next given what point one is at on the phase plane. The red point on the graph is representative of E_1 or the point $(0, U/d)$, which signifies orchard collapse. The blue point on the graph is representative of E_2 or $\left(\frac{U(A-B)-dBK_M}{aBK_M}, \frac{BK_M}{A-B}\right)$, which signifies orchard health and sustained fruit production. The nullclines are as defined in the legend, and the solid orange, red, and blue curves each represent a certain scenario: given any initial point $(F(0), M(0))$, the trajectory moves following the arrows and ends at an equilibrium. In the graph, all trajectories move towards E_2, signifying that E_2 is in fact locally asymptotically stable. The trajectory represented by red does not end at an equilibrium point, though if the x-axis was extended it would curve around and move towards E_2 as well. No trajectory moves towards E_1, consistent with its classification as a saddle point under the chosen parameter values used to derive this graph, though vectors along the y-axis can be seen moving towards from above and below.

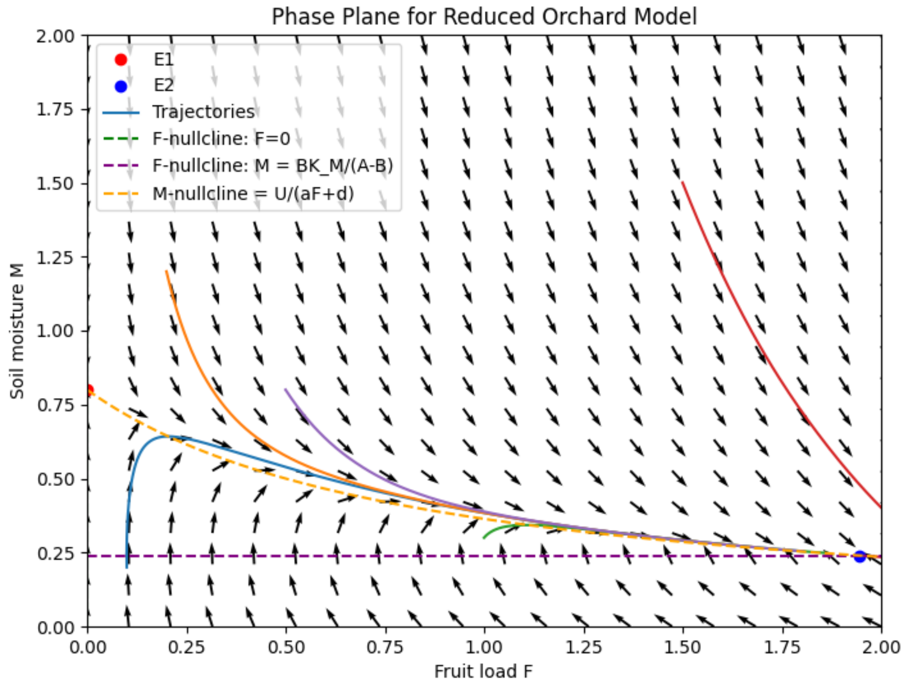

Figure 1: Phase plane of the reduced orchard model, showing the vector field, nullclines, equilibria, and representative trajectories.

4 Simulations

An advantage to the apple orchard yield model is the ability to simulate a variety of different real-world scenarios and investigate how they may impact fruit production, which by extension and subsequent modeling is useful for operations such as revenue forecasting (the scope of which is beyond this paper). The following numerical simulations revert back to the full model rather than the simplified one, reintroducing time-dependent external forcing to more realistically capture orchard behavior.

For the baseline simulation, shown in Figure 2, the model was solved numerically with initial conditions $F(0) = 0.30$ and $M(0) = 0.60$ (both standardized to fall between 0 and 1). Parameter values were selected to be $r = 0.10$, $K_M = 0.40$, $\alpha = 0.04$, $\beta = 0.03$, $K_L = 0.50$, $\gamma = 0.08$, $a = 0.08$, $b = 0.002$, and $c = 0.03$. Temperature thresholds were set to $T_{\min} = 55$ degrees Fahrenheit, $T_{\max} = 85$ degrees Fahrenheit, and T_f (freeze temperature) $= 28$ degrees Fahrenheit. Temperature, rainfall, irrigation, labor availability, and pest pressure were held at the standard conditions specified below:

$$
\begin{aligned}
T(t) &= 70 \text{ degrees Fahrenheit}, \\
R(t) &= 0.05 \text{ (rate; scaled water input/time)}, \\
I(t) &= 0.03 \text{ (rate; scaled water input/time)}, \\
L(t) &= 0.80 \text{ (standardized between 0 and 1)}, \\
P(t) &= 0.10 \text{ (standardized between 0 and 1)}.
\end{aligned}
$$

The specified conditions produce the following result:

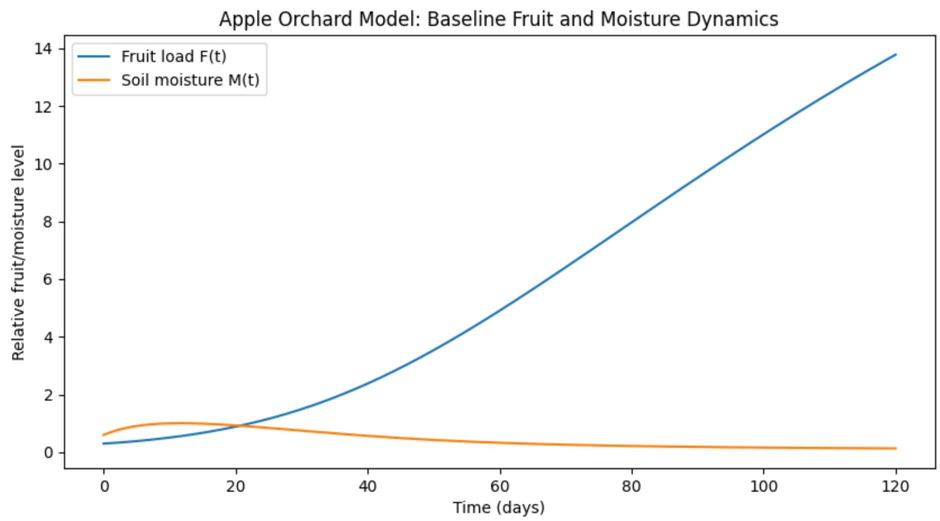

Figure 2: Baseline fruit and soil-moisture dynamics under standard orchard conditions.

In this simulation, over a timespan of 120 days fruit yield increases steadily at a nearly linear rate while soil moisture declines—though at a rate much slower than fruit production—and then eventually plateaus around day 100. This is not indicative of unideal conditions, but instead is due to the fact that as fruit grows more water from the soil is consumed. This simulation represents ideal orchard conditions for robust fruit production, though it should be noted that most likely there will come a time when more water input from rainfall and irrigation will be necessary to sustain fruit growth.

The next simulation involves the incorporation of a freeze event at days 40, 41, and 42 across the span of 120 days. All other inputs are held the same as they were in the prior simulation. The results are as follows:

In this simulation, shown in Figure 3, the freeze event—where temperature drops to 25 degrees Fahrenheit—that occurs over the span of days 40–42 causes a steep, near-linear drop in orchard productivity, and a nearly negligible boost in soil moisture due to decreased fruit production and therefore water usage. However, the orchard quickly recovers and continues on its path of steady fruit growth, in agreement with the stable equilibrium point that dictates that even amid small disturbances, positive fruit yield will prevail. Ultimately, the orchard is able to sustain a freeze event and adequately recover.

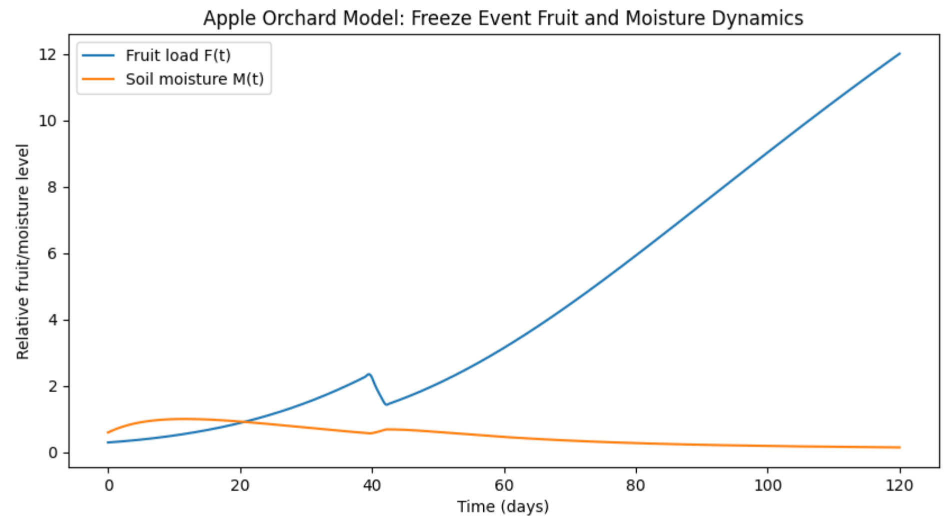

Figure 3: Fruit and soil-moisture dynamics under a short freeze event.

The final simulation, shown in Figure 4, involves a significant labor shortage where $L(t)$ is shifted from 0.80 to 0.30, and the penalty term, β, is shifted from 0.03 to 0.10. The shift in the penalty term represents higher significance of labor shortage due to a critical moment in the season such as thinning or harvest time. There is no longer a freeze event, and all other parameters are held the same as previously specified.

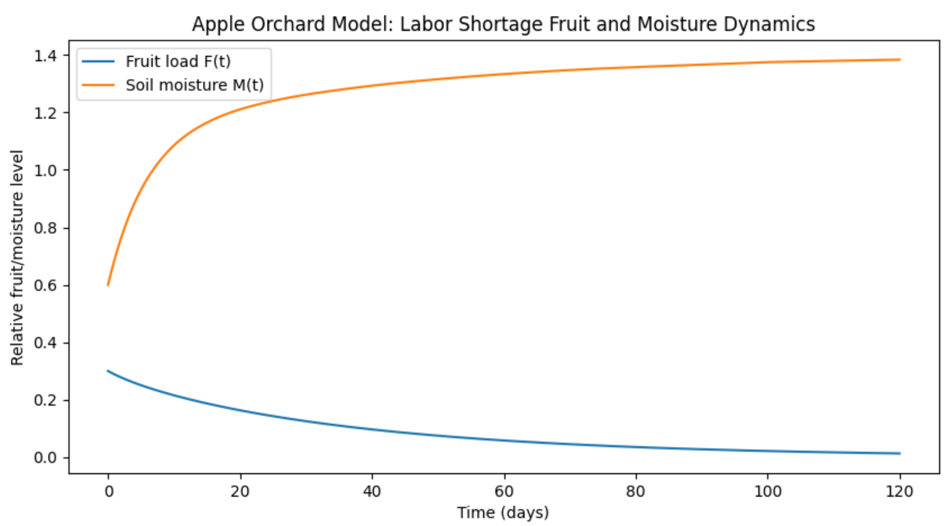

Figure 4: Fruit and soil-moisture dynamics under severe labor shortage during a critical point in the season.

In this simulation, fruit yield is significantly harmed, with an initial relatively steep drop and then movement towards a plateau, indicating that the orchard is not collapsing completely and instead is adjusting to a reduced but nonzero production equilibrium. Perhaps most noticeably, however, is the initial steep rise in soil moisture that then plateaus as well, though at a higher level than where it started. This can be explained, in essence, by labor shortage reducing fruit production, which in turn lowers water uptake. Rainfall and irrigation are now relatively larger than water consumption, causing a steep rise in soil moisture. Soil moisture then plateaus when water input and water loss come into balance again.

5 Conclusion

In summary, a model was derived using ordinary differential equations to demonstrate the relationship between fruit yield and soil moisture in a hypothetical apple orchard in the Pacific Northwest. Factors considered included labor availability, pest pressure, rainfall and irrigation, and temperature and freeze events, and their impact on the fruit yield and soil moisture dynamic. The model found a threshold condition for orchard viability, dictated by whether effective fruit growth supported by soil moisture exceeds loss aggregated by pressure from the aforementioned external influences. It revealed that water input from rainfall and irrigation heavily matters, and that not all conditions lead to sustainable production, one example being the simulation of a severe labor shortage during pruning or harvest season. Other simulations demonstrated a more robust orchard resilience, including a baseline scenario that supported increasing fruit yield and concurrent yet expected loss of soil moisture due to crop uptake, as well as the freeze event simulation that demonstrated that even a few days of harsh temperatures do not impede the orchard's ability to rebound and continue to grow in yield, in correlation with the finding of a locally asymptotically stable equilibrium point. By extension, the apple orchard is generally resilient in the face of small systemic shocks, and still moves towards positive equilibrium. Events such as heavy labor shortage at critical points in the season still have the ability to rock fruit production, even in the face of otherwise stable conditions. Ultimately, the model lends itself to farmers or investors alike in interpreting how extenuating circumstances may affect their crop and therefore forecasted revenue, leading to improved economic outcomes. For example, a predictable and impending labor shortage may be reason to divert investments or focus more heavily on the cultivation of another crop, boosting returns for either party. In the end, being prepared for all possible conditions in a time when national economic baseline is unsteady is best practice and ensures responsible decision making for all.

References

[1] Northwest Horticultural Council. *Apple Industry Facts.* `https://nwhort.org/industry-facts/apple-fact-sheet/`

[2] Conservation Resources. *Regenerative Agricultural Investments.* `https://www.conservationresources.net/regenerative-agricultural-investments`